Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Mechanics 1 – Elastic Collisions (1D)

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Two particles of masses 2 kg and 3 kg moving towards each other at $4\,\mathrm{m\,s^{-1}}$ and $2\,\mathrm{m\,s^{-1}}$ collide with $e = \frac{1}{2}$. Find the speeds after impact. **[5]**

2. A particle of mass m moving at u hits a stationary particle of mass $2m$ with $e = \frac{1}{3}$. Find the velocities after impact. **[5]**

3. Show that if $e = 1$ and masses are equal, the particles exchange velocities. **[3]**

4. A sphere hits a fixed vertical barrier normally with speed u and $e = \frac{3}{4}$. Find the rebound speed. **[2]**

5. Find the impulse on the barrier in the previous question if the mass is m. **[2]**

6. Two particles approach each other with equal speeds u and equal masses. Given e, find the speeds after impact. **[4]**

7. A particle of mass 4 kg at $5\,\mathrm{m\,s^{-1}}$ collides with a 2 kg particle at $1\,\mathrm{m\,s^{-1}}$ in the same direction. $e = 0.5$. Find final speeds. **[5]**

8. Find the loss in KE for the collision in Q1. **[4]**

9. A ball is dropped from 5 m onto a floor with $e = 0.8$. Find the speed just after the first bounce. **[4]**

10. How high does the ball in Q9 rebound? **[3]**

11. Three equal masses A, B, C in a line. A moving at u hits B at rest; $e = 1$. Then B hits C at rest; $e = 1$. Find final speeds. **[4]**

12. A particle hits a stationary particle of equal mass. Show that the paths after impact are perpendicular if $e = 1$ (oblique case preview). **[3]**

13. Write the two equations (momentum and restitution) for a general direct impact. **[3]**

14. A $1\,\text{kg}$ particle at $6\,\text{m}\,\text{s}^{-1}$ hits a $2\,\text{kg}$ particle at rest. After impact the $1\,\text{kg}$ particle is at rest. Find e. **[3]**

15. Find the range of possible values of e for a given collision scenario where both continue in the same direction. **[4]**

Section B – Hard (Q16–25)

16. Particles of masses m and $3m$ moving in the same direction at $4u$ and u collide with $e = \frac{1}{2}$. Find the velocities after impact and the impulse. **[6]**

17. A sphere is projected along a horizontal surface towards a fixed vertical wall. Coefficient of friction is μ, e with the wall. Discuss the motion. **[5]**

18. Two particles collide and the ratio of speeds of approach to separation is e. Derive the KE loss formula. **[5]**

19. A particle of mass m moving at u collides with a stationary particle of mass km. Find k in terms of e if the first particle stops. **[4]**

20. Successive bounces of a ball with e: show that the total distance travelled after first impact is $\frac{e^2 h}{1-e^2}$ (vertical drop h). **[5]**

21. Three particles masses $m, 2m, 3m$ in a line. First hits second with $e = \frac{1}{2}$, then second hits third with $e = \frac{1}{2}$. First has speed u, others at rest. Find final speeds. **[7]**

22. A bullet passes through a block (not embedding). Masses m and M, speeds u and v after. Find the impulse and discuss energy. **[5]**

23. Show that for equal masses and $e = 1$, KE is conserved in a direct collision. **[3]**

24. A particle hits a movable barrier of mass M with coefficient e. Set up equations. **[4]**

25. Find the value of e that maximises the transfer of KE to a stationary target of mass M from mass m. **[5]**

End of Worksheet • *Show all working clearly.*